Weichen Zhang

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Profile

Software Engineering graduate of Beihang University, with research experience in real-world AI and intelligent sensing under Professor Haixia Pan and Researcher Juntao Wang. Experience spans problem formulation, data construction, method design, experimental validation, and deployment, with a focus on connecting learning-based methods to reliable industrial systems. Following completion of the M.Eng., continued collaboration with my advisors has supported more method-oriented research beyond earlier industrial applications, including MoReAD (AAAI 2027 under review).

Education

Beihang University, Beijing, China — Sep. 2023 – Jan. 2026

Master of Engineering in Software Engineering — GPA: 3.75/4.00

Advisors: Professor Haixia Pan (Beihang University); Researcher Juntao Wang (China Aero-Polytechnology Establishment)

Beihang University, Beijing, China — Sep. 2018 – Jun. 2023

Bachelor of Engineering in Software Engineering — GPA: 3.73/4.00

Publications and Intellectual Property

- **ZHANG W**, FENG X, LIU Y, PAN H. MoReAD: Cross-Representation Mixture-of-Experts and Conditional Recovery for Unsupervised Industrial Anomaly Detection [C/OL]. *AAAI 2027 Conference Submission*, 2026 (under review).
- **ZHANG W**, XU R, DU Y, WANG J. Aircraft Frost Detection Based on Laser Thickness and Visual Analysis: China, ZL 2025 1 0773975.6 [P]. 2026-07-24. (Granted; first inventor.)
- **ZHANG W**, ZHENG B, MA M, ZHOU Y. High-Precision Visual Positioning System Based on Salient Object Detection, V1.0: China, 2024SR0923296 [CP]. 2024-07-03. (Registered.)

Research Experience

MoReAD: Cross-Representation MoE for Industrial Anomaly Detection — Jan. 2026 – Jul. 2026

PyTorch, Mixture-of-Experts, Knowledge Distillation, Self-supervised Learning, Unsupervised anomaly detection AAAI under review

- Studied how incomplete representations and ambiguous recovery can make normal variation resemble defects, and developed a feature-reconstruction framework with CR-MoE to distill training-only EUPE expert evaluation into a lightweight DINO router.
- Developed CBTM for sample-conditioned low-rank feature recovery without anomaly-leaking shortcuts, while retaining a DINO-only inference path after removing auxiliary branches.
- Achieved state-of-the-art anomaly localization across five benchmarks, improving P-AUPRO@0.05 by an average of 2.9 percentage points over the strongest baseline on each dataset.

Multimodal Fusion for Airport Runway FOD Detection — Jan. 2023 – Jun. 2023

PyTorch, YOLO, Embedded C++, Millimeter-Wave Radar, Multimodal Fusion

- Reproduced a published baseline without source code, implementing its model, loss functions, training strategy, and evaluation pipeline.
- Developed Continental radar software in Python and a TI acquisition module in embedded C++; completed camera calibration, radar–camera mapping, and radar–RGB dataset construction.
- Designed and deployed an all-weather small-object detector that fused radar projections with RGB features through spatial and channel attention, and validated it against unimodal methods.

Self-Supervised Aircraft Skin Anomaly Detection — Feb. 2024 – Jan. 2026

PyTorch, Unsupervised Anomaly Detection, Frequency Modeling, Cross-Attention

- Built and field-deployed an acquisition prototype and dataset of 1,258 high-resolution (2448×2048) aircraft skin images; designed low-frequency-first decomposition and cross-frequency attention.
- Designed masked pseudo-anomalies and latent perturbation to address defect scarcity and reconstruction bias, achieving 94.7% image AUROC, 92.6% pixel AUROC, and 89.7% AUPRO.

Selected Applied Research and Engineering

Aircraft Surface Frost Detection Using Laser Thickness Measurement and Visual Analysis

Laser Sensing, CNN Feature Matching, Anomaly Detection, Score Fusion, modbus TCP Granted patent; first inventor

- Developed the control, communication, and system software and operator interface for synchronized laser, monochrome RGB camera, and environmental-sensor acquisition.
- Personally collected, cleaned, and annotated the dataset; implemented modified Beer–Lambert estimation, CNN feature matching, and score fusion, achieving validation MSE 0.0024 versus 0.0084 laser-only and 0.0221 vision-only.

Industrial CT-Based Aircraft Blade Metrology and Quality Control Platform

TransUNet, Keypoint Detection, Segmentation, OpenCV, PyQt5, Industrial CT

- Designed and implemented the PyQt5 platform for CT image management, model inference, dimensional measurement, tolerance assessment, report generation, and electronic-signature integration; deployed at China Aero-Polytechnology Establishment.
- Combined an existing TransUNet segmenter with task-specific keypoint detection and classical-CV mask refinement, then integrated automated measurement with traceable manual reinspection.

End-to-End Industrial Vision-Guided Robotic Grasping Platform

PyTorch, OpenCV, Calibration, Flask, PyQt5, Modbus TCP Software copyright registered

- Led model training/deployment and software integration across image acquisition, intrinsic/hand–eye calibration, Flask/PyQt applications, and Modbus robot control; deployed at JOMOO.
- Trained and deployed a hybrid ResNet18–T2T-ViT model with multi-scale fusion (over 93% mIoU), implemented contour-based pose extraction, and collaborated with JOMOO hardware engineers on closed-loop grasp verification.

Six-Camera Real-Time Bubble Inspection Platform for PTFE Tubes

Multi-Camera Acquisition, Producer–Consumer Queues, CUDA Streams Using in Shanghai plastic Research Institution

- Trained and deployed the YOLO bubble-defect detector and designed the six-camera acquisition, inference, and result-aggregation software pipeline for a single RTX 3060.
- Implemented producer–consumer buffering, CUDA Streams, and dynamic cross-camera batching to decouple capture from inference and support stable continuous production inspection.

Additional Engineering: Implemented a Java/ANTLR SysY-to-LLVM compiler; developed Spring Boot/Vue services with Elasticsearch/Redis search and YOLO–ByteTrack personnel tracking dashboards.

Industry Experience

China Aero-Polytechnology Establishment, Beijing, China — Jul. 2024 – Sep. 2025

Software Engineer Intern (Computer Vision) Department of Manufacturing Standardization and Quality Technology

Served as technical or software lead for aviation visual inspection systems, covering requirements, data acquisition, model development, application engineering, device integration, on-site deployment, and operational support.

Additional Academic Experience

Research Proposal Development: Contributed to a civil aviation research fund proposal on aircraft frost monitoring, a National Key R&D Program proposal, and NSFC proposals on AIGC-based medical imaging and digital-twin industrial management; drafted objectives, technical roadmaps, innovation statements, and implementation plans.

Honors and Awards

Academic Excellence Scholarships (3 awards); Academic Competition Scholarship	2018–2023
Graduate Excellence Scholarships: Qihang (1 award) and Linghang (2 awards)	2023–2025

Skills

- **Computer Vision Methods:** object detection, semantic segmentation, keypoint localization, anomaly localization
- **Representation and Anomaly Learning:** representation learning, feature reconstruction, self-supervised learning, feature matching, knowledge distillation, attention-based modeling
- **Multimodal Sensing:** radar–RGB fusion, laser sensing, camera calibration, attention-based fusion, industrial cameras
- **Deployment and Systems:** model deployment, CUDA Streams, multi-camera inference, PyQt5, Flask/REST, Modbus TCP, Spring Boot, Vue.js
- **Programming and Data:** Python, Java, C/C++, JavaScript, SQL, MySQL, Elasticsearch, Redis; embedded C++ and Java/ANTLR
- **Languages, Certification and Interests:** Chinese (native); English (IELTS 7.0); PADI Advanced Open Water Diver; fitness, cycling, scuba diving